

A Smooth Penetration Depth for Point–Loop Contact

Half-Plane Log-Sum-Exp Softmin, Hertzian Repulsion, and Adhesion

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1 Setting and requirements

Let $\Omega \subset \mathbb{R}^2$ be the open region bounded by a simple closed polygonal loop $\partial\Omega$ with vertices $v_1, \dots, v_N$ in counterclockwise order, edges $e_i = v_{i+1} - v_i$ (indices mod N), unit tangents $\hat{t}_i = e_i/|e_i|$, and outward unit normals $n_i = J\hat{t}_i$ with

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J^2 = -I, \quad Jn_i = -\hat{t}_i. \quad (1)$$

A point x interacts with the loop through a penetration depth $h(x)$ that is positive when $x \in \Omega$ and negative outside. We require of the surrogate depth h_ε :

(i) *analyticity*. $h_\varepsilon \in C^\infty(\mathbb{R}^2)$, and ideally real-analytic, so that Newton solves and quasistatic branch-following see no artificial nonsmoothness anywhere, not merely away from a measure-zero set.

(ii) *closed-form derivatives*. Gradient and Hessian available algebraically, without quadrature or finite differencing.

(iii) *metric fidelity*. $h_\varepsilon \approx \text{dist}(x, \partial\Omega)$ with $|\nabla h_\varepsilon| \approx 1$ over the physically relevant range of penetrations, so the contact law retains its intended force–displacement relation.

(iv) *exact sign*. The zero set of h_ε is a well-defined closed curve, so contact detection is unambiguous.

(v) *conservation*. Exact translational and rotational invariance of the resulting energy, so forces and torques on the loop’s vertices sum to zero by construction rather than to within truncation error.

Section 2 shows that (i) and (iii) cannot both hold exactly. Sections 3–8 construct the surrogate that concedes (iii) only where it is physically irrelevant, and show that the concession is quantified by a single length ε which simultaneously sets the corner rounding radius, the medial-axis smearing width, and the magnitude of the Hessian. Sections 9–13 then add adhesion, which introduces a second length λ and an energy W , and which turns out to repair rather than aggravate the conditioning of the contact Hessian.

2 The obstruction: no smooth exact depth exists

Let ϕ denote the signed distance function, $\phi(x) = \text{dist}(x, \partial\Omega)$ for $x \in \Omega$ and $\phi(x) = -\text{dist}(x, \partial\Omega)$ otherwise. Where ϕ is differentiable it satisfies the eikonal equation $|\nabla\phi| = 1$, since moving along the ray to the nearest boundary point changes ϕ at unit rate. But Ω is bounded, so ϕ attains

an interior maximum equal to the inradius r_{in} , and at an interior maximum $\nabla\phi = 0$. Hence ϕ cannot be C^1 on all of Ω ; the obstruction is topological, not an artifact of the corners.

The non-differentiability locus is the **medial axis** $\mathcal{M}$, the set of points possessing two or more nearest boundary points. For any non-degenerate loop $\mathcal{M}$ is a connected skeleton of codimension one, and $\nabla\phi$ jumps across it by the difference of the competing unit normals. There is a second, weaker defect: even where ϕ is C^1 , it is generally not C^2 . Outside a convex Ω the nearest point is unique and ϕ is C^1 globally, but across the ray normal to a vertex the level sets switch from straight lines to circular arcs, so $\nabla^2\phi$ jumps from 0 to $O(1/\text{dist})$. Inside, the same switching occurs across the boundaries between the Voronoi cells of edges and of vertices.

Any smooth surrogate must therefore break either the eikonal property or the exactness of the zero set somewhere, and the only real design decision is where to spend the error. For contact physics the answer is dictated by scale. The medial axis sits at penetration depths of order r_{in} , which for any sensible soft-particle model is far beyond the regime the potential is meant to describe; the vertex Voronoi boundaries reach all the way to the boundary and are unavoidable. So the correct trade is: preserve $|\nabla h_\varepsilon| = 1$ to exponential accuracy in the shallow-penetration regime, absorb all error into an ε -collar of $\mathcal{M}$ and an ε -neighborhood of the vertices, and accept an $O(\varepsilon)$ inward displacement of the zero set as the definition of the particle's shape.

This last point deserves emphasis. It is tempting to build a smooth approximation to a polygon and then treat the discrepancy as an error to be driven to zero. The more useful reading is that the smooth object *is* the model: h_ε defines a particle with analytically rounded corners, and the polygon is merely the sharp- ε limit that motivated it. Under that reading ε is a physical shape parameter with a mechanical role, not a numerical regulator.

3 Half-plane representation and exactness in the interior

Take Ω convex for now; Section 15 treats the general case by composition. Define for each edge the affine function

$$\ell_i(x) = n_i \cdot (v_i - x) = c_i - n_i \cdot x, \quad c_i = n_i \cdot v_i = n_i \cdot v_{i+1}, \quad (2)$$

the equality of the two expressions for c_i following from $n_i \perp e_i$. Then $\ell_i(x)$ is the signed distance from x to the full line L_i containing edge i , positive on the interior side, and $\nabla\ell_i = -n_i$ with $|\nabla\ell_i| = 1$ and $\nabla^2\ell_i = 0$. Convexity gives $\Omega = \bigcap_i \{\ell_i > 0\}$.

Lemma 1 (Interior exactness). *For convex Ω and every $x \in \overline{\Omega}$,*

$$\text{dist}(x, \partial\Omega) = \min_{1 \leq i \leq N} \ell_i(x). \quad (3)$$

Proof. Fix i . The orthogonal projection of x onto L_i is $q_i = x + \ell_i(x) n_i$, at distance $\ell_i(x)$ from x , and $\ell_i(q_i) = 0$ so $q_i \notin \Omega$. The segment $[x, q_i]$ therefore joins a point of $\overline{\Omega}$ to a point of the complement of Ω and must meet $\partial\Omega$, whence $\text{dist}(x, \partial\Omega) \leq \ell_i(x)$. Taking the minimum over i gives $\text{dist}(x, \partial\Omega) \leq \min_i \ell_i(x)$.

Conversely let $p \in \partial\Omega$ realize the distance, $|x - p| = \text{dist}(x, \partial\Omega)$. Since the closed edges cover $\partial\Omega$, there is an index j with $p \in L_j$, and the distance from x to L_j cannot exceed the distance to a particular point of L_j , so $\min_i \ell_i(x) \leq \ell_j(x) \leq |x - p|$. $\square$

Outside Ω exactness fails: in the normal fan beyond a vertex the true distance is to the vertex itself while $\min_i \ell_i$ returns the negative of the larger line distance, so $\min_i \ell_i \geq \phi$ with

strict inequality. The sign is nonetheless exact everywhere, $\min_i \ell_i > 0$ iff $x \in \Omega$, which is all that a repulsive-only contact law requires: the exterior magnitude never enters the energy.

The choice of the affine ℓ_i over the distance-to-*segment* functions

$$d_i(x) = |x - v_i - \text{clamp}_{[0,1]}(\hat{t}_i \cdot (x - v_i)/|e_i|) e_i| \quad (4)$$

is the crux of the construction. Both satisfy $\phi = \min_i(\pm)$ appropriately, but each d_i is itself only C^1 , with a Hessian that jumps across the two lines normal to the segment at its endpoints. Smoothing the minimum does nothing about that: the result is C^1 but not C^2 , on a locus that extends to arbitrarily small penetration. The affine ℓ_i are entire, so *every* defect of the exact distance function — gradient jumps on the medial axis and Hessian jumps on the Voronoi walls alike — has been pushed into the single operation of taking a minimum. It then suffices to smooth that one operation.

4 The log-sum-exp soft depth

Define, for a length scale $\varepsilon > 0$,

$$h_\varepsilon(x) = -\varepsilon \log \sum_{i=1}^N e^{-\ell_i(x)/\varepsilon} \quad (5)$$

the softmin of the half-plane distances. Since each summand is entire and strictly positive, the sum is a strictly positive entire function and h_ε is real-analytic on all of $\mathbb{R}^2$; requirement (i) holds in the strong sense, with no exceptional set.

Proposition 1 (Bounds). *Write $\ell_{(1)} \leq \ell_{(2)} \leq \dots$ for the sorted values and $\Delta = \ell_{(2)} - \ell_{(1)} \geq 0$. Then*

$$\ell_{(1)} - \varepsilon \log N \leq h_\varepsilon \leq \ell_{(1)} - \varepsilon \log(1 + (N-1)e^{-\Delta/\varepsilon}) \leq \ell_{(1)}. \quad (6)$$

Proof. Each of the N terms is at most $e^{-\ell_{(1)}/\varepsilon}$ and at least one equals it, so $e^{-\ell_{(1)}/\varepsilon} \leq \sum_i e^{-\ell_i/\varepsilon} \leq N e^{-\ell_{(1)}/\varepsilon}$; applying $-\varepsilon \log(\cdot)$ reverses and scales. For the sharper form, bound each of the $N-1$ subdominant terms by $e^{-\ell_{(2)}/\varepsilon}$, giving $\sum_i e^{-\ell_i/\varepsilon} \leq e^{-\ell_{(1)}/\varepsilon} (1 + (N-1)e^{-\Delta/\varepsilon})$. $\square$

Thus h_ε never exceeds the exact depth, the global deficit is at most $\varepsilon \log N$, and the local deficit is bounded by $\varepsilon(N-1)e^{-\Delta/\varepsilon}$, exponentially small wherever a single face is closest by a margin of several ε . The set where the deficit is not negligible is precisely the ε -collar of $\{\Delta \lesssim \varepsilon\}$, which for interior points is the medial axis and for near-boundary points is the vertex neighborhoods. The error has landed exactly where Section 2 argued it should.

Since h_ε is a softmin of affine functions it is **concave**: $\log \sum \exp$ is convex and nondecreasing in each argument, $-\ell_i/\varepsilon$ is affine, and the sign flip in (5) converts convexity to concavity. Concavity of the depth is not a defect but a statement that the shape is convex, and it has a direct mechanical consequence developed in Section 8.

5 Derivatives in closed form

Introduce the softmin weights

$$w_i(x) = \frac{e^{-\ell_i(x)/\varepsilon}}{\sum_j e^{-\ell_j(x)/\varepsilon}}, \quad w_i > 0, \quad \sum_i w_i = 1, \quad (7)$$

and the weighted mean normal $\bar{n} = \sum_i w_i n_i$. Differentiating (5),

$$\nabla h_\varepsilon = \sum_i w_i \nabla \ell_i = - \sum_i w_i n_i = -\bar{n}. \quad (8)$$

For the Hessian, note $\nabla \log Z = -\varepsilon^{-1} \sum_j w_j \nabla \ell_j = \varepsilon^{-1} \bar{n}$ with $Z = \sum_j e^{-\ell_j/\varepsilon}$, so

$$\nabla w_i = w_i \left(-\frac{\nabla \ell_i}{\varepsilon} - \nabla \log Z \right) = \frac{w_i}{\varepsilon} (n_i - \bar{n}), \quad (9)$$

and therefore

$$\nabla^2 h_\varepsilon = - \sum_i n_i \otimes \nabla w_i = -\frac{1}{\varepsilon} \left[\sum_i w_i n_i n_i^\top - \bar{n} \bar{n}^\top \right] = -\frac{1}{\varepsilon} \text{Cov}_w(n), \quad (10)$$

manifestly symmetric, and negative semidefinite as concavity requires, since a covariance matrix is positive semidefinite. All of (8)–(10) cost $O(N)$ evaluations of an exponential and no transcendental inversions; requirement (ii) holds.

Two identities make the geometric content explicit. First, using $\sum_i w_i = 1$, $|n_i| = 1$, and $n_i \cdot n_j = 1 - \frac{1}{2}|n_i - n_j|^2$,

$$|\nabla h_\varepsilon|^2 = |\bar{n}|^2 = \sum_{i,j} w_i w_j n_i \cdot n_j = 1 - \frac{1}{2} \sum_{i,j} w_i w_j |n_i - n_j|^2 \leq 1. \quad (11)$$

The soft depth is thus 1-Lipschitz, and its eikonal defect equals exactly the weighted mean-square spread of the competing outward normals. Where one face dominates, w concentrates and $|\nabla h_\varepsilon| = 1$ to exponential accuracy; requirement (iii) holds in the shallow regime. Second, taking the trace of (10) and using $\text{tr Cov}_w(n) = \sum_i w_i |n_i|^2 - |\bar{n}|^2 = 1 - |\bar{n}|^2$,

$$\varepsilon \Delta h_\varepsilon = -(1 - |\nabla h_\varepsilon|^2), \quad \text{i.e.} \quad \boxed{|\nabla h_\varepsilon|^2 - \varepsilon \Delta h_\varepsilon = 1} \quad (12)$$

holds identically at every point of $\mathbb{R}^2$. Curvature of h_ε and departure from the eikonal law are the same quantity: the surrogate is stiffest exactly where it is least metrically faithful, and both are localized to the same ε -collar. Equation (12) also supplies a machine-precision unit test (Section 17).

6 Why this construction is canonical

Equation (12) is the viscosity regularization of the eikonal equation, and it identifies (5) as the exact solution of a linear problem rather than an ad hoc smoothing. Substituting $u = e^{-h_\varepsilon/\varepsilon}$, so that $\nabla u = -u \nabla h_\varepsilon/\varepsilon$ and $\Delta u = u(|\nabla h_\varepsilon|^2/\varepsilon^2 - \Delta h_\varepsilon/\varepsilon)$, turns (12) into

$$\varepsilon^2 \Delta u = u, \quad (13)$$

the screened Poisson (modified Helmholtz) equation. The Cole–Hopf substitution is exact, not asymptotic. Conversely, each plane wave $u_i = e^{-\ell_i/\varepsilon}$ solves (13) because ℓ_i is affine with $|\nabla \ell_i| = 1$ and $\Delta \ell_i = 0$; by linearity so does any positive combination, and (5) is simply $-\varepsilon \log$ of the particular combination that reproduces the polygon’s faces.

This is the Varadhan characterization of distance: solving $\varepsilon^2 \Delta u = u$ with $u = 1$ on $\partial\Omega$ and forming $-\varepsilon \log u$ recovers $\text{dist}(\cdot, \partial\Omega)$ as $\varepsilon \rightarrow 0$, with the logarithm converting the exponentially

small tails of a diffusive kernel into a metric. The half-plane sum (5) is the closed-form instance of that construction for a convex polygon. Two consequences follow. The $O(\varepsilon)$ deviations are not arbitrary: they are the unique smoothing generated by linear screening, which is why the resulting derivatives are as clean as (8)–(10). And the family is closed under superposition, so richer shapes can be built by adding solutions — point sources $K_0(|x - p|/\varepsilon)$ for rounded vertices, for instance — without leaving the class in which (12) holds.

7 Geometry of the zero set

Proposition 1 gives $h_\varepsilon \leq \ell_{(1)}$, so the region $\Omega_\varepsilon = \{h_\varepsilon > 0\}$ is a proper subset of Ω : the zero level set is an inward offset of $\partial\Omega$. On the interior of an edge, far from both vertices, the offset is $\varepsilon(N-1)e^{-\Delta/\varepsilon}$ and utterly negligible. At a corner it is not. Let two edges meet with interior angle θ , so their outward normals subtend $\pi - \theta$; write $\alpha = (\pi - \theta)/2$ for the half-angle, so $n_{1,2} = (\cos \alpha, \pm \sin \alpha)$ in a frame whose first axis bisects them. On the bisector $w_1 = w_2 = \frac{1}{2}$ and all other weights are negligible, so $h_\varepsilon = \ell - \varepsilon \log 2$ with $\ell = \ell_1 = \ell_2$, and the zero set crosses the bisector at

$$\ell_{\text{apex}} = \varepsilon \log 2, \quad (14)$$

independent of the corner angle. The rounding is smooth rather than circular but has a well-defined radius. With $t = \hat{y}$ tangent to the level set on the bisector, $t^\top \text{Cov}_w(n) t = \sin^2 \alpha$ and $|\nabla h_\varepsilon| = |\bar{n}| = \cos \alpha$, so the level-set curvature $\kappa = -t^\top \nabla^2 h_\varepsilon t / |\nabla h_\varepsilon|$ gives

$$R_{\text{corner}} = \frac{1}{\kappa} = \varepsilon \frac{\cos \alpha}{\sin^2 \alpha} = \varepsilon \frac{\sin(\theta/2)}{\cos^2(\theta/2)}. \quad (15)$$

For a right angle $R_{\text{corner}} = \sqrt{2}\varepsilon$; as $\theta \rightarrow \pi$ the radius diverges and the boundary flattens, as it must; as $\theta \rightarrow 0$ the radius collapses, the level set turning through a large angle within a shrinking region while the apex offset stays at $\varepsilon \log 2$. Sharp spikes are thus the expensive feature, and (15) is the quantitative statement of requirement (iv): the zero set is a smooth closed curve for any ε , and ε is the corner radius up to the angular factor. This is the same object as an explicitly constructed rounded polygon, obtained analytically instead of by fiat, with one parameter controlling corner rounding, medial-axis smearing, and Hessian magnitude at once.

8 Hertzian contact energy

Take the pair energy

$$E(x) = \frac{2}{5} k [h_\varepsilon(x)]_+^{5/2}, \quad [z]_+ = \max(z, 0), \quad (16)$$

with stiffness k of dimension energy/length^{5/2}. The Hertzian exponent is chosen for regularity at first contact, and the point is worth making precisely because it is independent of everything above: analyticity of h_ε does not by itself make E smooth. Writing $E = \varphi(h_+)$ with $\varphi(z) = \frac{2}{5} k z^{5/2}$, one has $\varphi'(z) = k z^{3/2}$ and $\varphi''(z) = \frac{3}{2} k z^{1/2}$, both vanishing as $z \rightarrow 0^+$, so $\varphi \in C^2(\mathbb{R})$ when extended by zero; only the third derivative $\frac{3}{4} k z^{-1/2}$ diverges. A harmonic law $\frac{1}{2} k h_+^2$ has $\varphi'' = k$ on one side and 0 on the other and is only C^1 , which shows up as a jump in the stiffness matrix every time a contact is created or destroyed — exactly the events that dominate an AQS

trajectory. Any exponent ≥ 3 would also serve; $5/2$ is the smallest that delivers C^2 and carries the Hertz interpretation.

Composing with (8) and (10), and abbreviating $h = h_\varepsilon$ and $\bar{n} = \sum_i w_i n_i$, the force on the point and the contact stiffness are

$$f = -\nabla_x E = -k h_+^{3/2} \nabla h = k h_+^{3/2} \bar{n}, \quad (17)$$

$$\nabla_x^2 E = \frac{3}{2} k h_+^{1/2} \bar{n} \bar{n}^\top - \frac{k h_+^{3/2}}{\varepsilon} \text{Cov}_w(n). \quad (18)$$

Three features of (17)–(18) matter for the physics.

The contact normal is the weighted mean $\bar{n}/|\bar{n}|$ of the competing face normals, which rotates smoothly as the point slides past a corner instead of jumping. The force magnitude, however, is $kh^{3/2}|\bar{n}| = kh^{3/2} \cos \alpha$ at a corner rather than $kh^{3/2}$: contact against a rounded corner is softer than against a face by the factor $|\bar{n}| \leq 1$ of (11). This is a genuine property of the smoothed shape, not an error, and it is the mechanical signature of ε .

The second term of (18) is negative semidefinite. Along a direction t tangent to the level set of h the first term vanishes and the eigenvalue is $-kh^{3/2} t^\top \text{Cov}_w(n) t / \varepsilon < 0$ wherever the normals disagree. The contact block is therefore *indefinite* near corners and near the medial axis. This too is physical: for any smooth convex indenter the transverse stiffness carries a term $-f/R$ with R the local radius of curvature, and comparing with (15) identifies precisely that geometric softening here. The consequence for the solver is that the assembled Hessian cannot be assumed positive definite; Newton steps require a modified factorization, a trust region, or a shifted preconditioner.

Comparing magnitudes of the two terms in (18) gives the crossover $\frac{3}{2}kh^{1/2} \sim kh^{3/2}/\varepsilon$, i.e. $h \sim \frac{3}{2}\varepsilon$. Contacts shallower than ε are dominated by the elastic term and produce positive-definite blocks; contacts substantially deeper than ε are dominated by the geometric term. For the purely repulsive law this forces ε to be comparable to or larger than the deepest expected overlap if positive-definite blocks are wanted, which is an uncomfortable constraint because ε is simultaneously a shape parameter. Section 12 shows that adding adhesion removes the constraint: the geometric term is proportional to $\varphi'(h)$, which vanishes identically at the adhesive equilibrium indentation, so contacts sitting near equilibrium are well conditioned regardless of how small ε is.

9 Adhesion: why a target penetration depth is the wrong primitive

The cheapest route to adhesion is to displace the minimum of the contact potential to a preferred overlap $h_0 > 0$, replacing (16) by $\frac{2}{5}k|h_\varepsilon - h_0|^{5/2}$. The underlying intuition is sound: an adhesive contact does equilibrate at nonzero indentation under zero applied load, and that is the content of the JKR solution. The implementation is not, for four reasons, and three of them are structural rather than cosmetic.

The bond cannot break. The energy grows without bound as the point separates, so the construction is a confining spring with infinite tensile strength rather than an adhesive contact with a finite work of separation. Truncating it at some range restores breakability but reinstates exactly the discontinuity in φ'' at contact loss that the Hertzian exponent was chosen in Section 8 to eliminate, and now at the more damaging location, since bond rupture is a far more frequent event along an AQS trajectory than first touch.

The work of adhesion and the elastic modulus are not independently adjustable. A single stiffness k and offset h_0 fix both, with the depth of the well scaling as $kh_0^{5/2}$. Adhesion is then useless as a control axis, because sweeping it necessarily sweeps the modulus, and the brittle–ductile axis one is trying to construct is confounded with a trivial rescaling of stress.

Adhesion is an interfacial quantity and h_ε is not. The physical adhesion energy of a contact scales with the length of shared interface, and that length depends strongly on contact *type*: a face-on-face contact between two polygons has a chord of order the edge length and nearly independent of indentation, while a vertex-on-face contact has a chord that vanishes with indentation. A rule that depends only on h_ε assigns these the same adhesion at equal depth, which erases the distinction that generates contact-chord bistability. Section 13 restores it.

Finally, a smooth single-valued well is a conservative potential and by itself produces no hysteresis: loading and unloading traverse the same path. Hysteretic bonding requires a non-convex energy with a fold, and if one is going to engineer non-convexity deliberately then there is a construction, given next, that supplies it together with the correct parameter count and requires no cutoff at all.

10 The adhesive energy

Retain the Hertzian repulsion and add a separate adhesive term built from the *saturating* mollifier,

$$E(x) = \frac{2}{5}k[h_\varepsilon(x)]_+^{5/2} - W\chi_\lambda(h_\varepsilon(x)), \quad \chi_\lambda(h) = \frac{1}{2} \left(1 + \frac{h}{\sqrt{h^2 + \lambda^2}} \right). \quad (19)$$

Here λ is an adhesive range and W an energy. The function χ_λ is the Plummer-mollified Heaviside: it increases monotonically from 0 at $h \rightarrow -\infty$ to 1 at $h \rightarrow +\infty$ with transition width λ , and is real-analytic on all of $\mathbb{R}$.

Its use here is the exact converse of its rejection in Section 16. What disqualifies χ_λ as a depth is that it saturates, carrying no information beyond $|h| \sim \lambda$. What qualifies it as an adhesion is that it saturates: once contact is established the bond energy must stop growing, and the interaction must have finite total strength. The two mollifiers are complementary, log-sum-exp for the metric and Plummer for the membership, each useless in the other's role, and they share primitives so the kernels can be fused.

Because χ_λ is entire, the adhesive term requires no positive part, no cutoff, and no case analysis; it contributes no discontinuity to any derivative anywhere. Its derivatives are elementary:

$$\chi'_\lambda(h) = \frac{\lambda^2}{2(h^2 + \lambda^2)^{3/2}}, \quad \chi''_\lambda(h) = -\frac{3\lambda^2 h}{2(h^2 + \lambda^2)^{5/2}}, \quad \chi'''_\lambda(h) = -\frac{3\lambda^2(\lambda^2 - 4h^2)}{2(h^2 + \lambda^2)^{7/2}}. \quad (20)$$

Work of adhesion. The tensile force conjugate to depth is $W\chi'_\lambda(h)$, and

$$\int_{-\infty}^{\infty} W\chi'_\lambda(h) dh = W[\chi_\lambda]_{-\infty}^{\infty} = W, \quad (21)$$

so W is exactly the work required to separate the bond from full engagement to infinity, independently of k and λ . Requirement (iii) of Section 9 is met: W , λ and k are three independent parameters with three distinct physical meanings.

Tensile strength. Since χ''_λ vanishes only at $h = 0$, the adhesive force is maximal there,

$$F_{\text{pull}} = W\chi'_\lambda(0) = \frac{W}{2\lambda}, \quad (22)$$

which is the force-controlled pull-off strength of a single bond. Note that W and λ enter the strength only through W/λ while the work of separation is W alone; strength and toughness are therefore separately tunable, which is the minimum requirement for a model that is supposed to interpolate between brittle and ductile response.

Emergent equilibrium indentation. Write $\varphi(h)$ for the scalar law in (19), so that $E = \varphi(h_\varepsilon)$ with

$$\varphi'(h) = k[h]_+^{3/2} - W\chi'_\lambda(h), \quad \varphi''(h) = \frac{3}{2}k[h]_+^{1/2} - W\chi''_\lambda(h). \quad (23)$$

The equilibrium indentation solves $\varphi'(h^*) = 0$. Setting $\eta = h^*/\lambda$ and clearing denominators,

$$k\lambda^{3/2}\eta^{3/2} = \frac{W}{2\lambda(1+\eta^2)^{3/2}} \iff \boxed{\eta^3 + \eta = \mu, \quad \mu \equiv \left(\frac{W}{2k\lambda^{5/2}}\right)^{2/3}}. \quad (24)$$

The preferred overlap is thus not an input but the root of a depressed cubic in a single dimensionless group, with the Cardano form

$$\eta(\mu) = \sqrt[3]{\frac{\mu}{2} + \sqrt{\frac{\mu^2}{4} + \frac{1}{27}}} + \sqrt[3]{\frac{\mu}{2} - \sqrt{\frac{\mu^2}{4} + \frac{1}{27}}}, \quad (25)$$

the second radicand being negative and the real cube root intended. The discriminant $\mu^2/4 + 1/27$ is strictly positive, so the root is unique for every $\mu > 0$: the contact has exactly one equilibrium indentation, never several.

The control parameter. The group μ in (24) is the ratio of the elastic indentation produced by the peak adhesive force to the range of that force, and is therefore a Tabor parameter for this contact law. Its two limits are the standard ones. For $\mu \ll 1$, $\eta \simeq \mu - \mu^3$ and $h^* \simeq (W/2k\lambda)^{2/3}$: adhesion acts as a nearly constant external load on an otherwise Hertzian contact, depends strongly on λ , and deforms it little. This is the DMT regime, weakly cohesive and ductile. For $\mu \gg 1$, $\eta \simeq \mu^{1/3}$ and $h^* \simeq (W\lambda^{1/2}/2k)^{1/3}$, only weakly dependent on the range: adhesion dominates the contact geometry and the equilibrium becomes a property of W and k alone. This is the JKR regime, strongly cohesive, with large stored energy released on rupture, and brittle. Sweeping μ at fixed k is therefore a physically interpretable brittle–ductile axis, and it is a cleaner control knob than a disorder distribution parameter because it does not simultaneously alter the statistics of the quenched landscape.

Binding energy. The depth of the well is $-\varphi(h^*) = W\chi_\lambda(h^*) - \frac{2}{5}k(h^*)^{5/2}$, strictly less than W because part of the adhesive gain is spent on elastic compression. The distinction matters when converting a measured cohesion energy into a value of W ; in the DMT limit the two coincide to $O(\mu^{5/3})$, in the JKR limit they do not.

11 Bond stability and the hysteron threshold

For $h < 0$ the Hertzian term is absent and $\varphi'' = -W\chi_\lambda'' = \frac{3}{2}W\lambda^2h(h^2 + \lambda^2)^{-5/2} < 0$. The entire separated branch has negative stiffness, so a bond under tension is unstable unless stabilized by its surroundings. Quantitatively, from (20) the extrema of χ_λ'' are the zeros of χ_λ''' , namely $h = \pm\lambda/2$ exactly, and evaluating there gives

$$\min_h \varphi''(h) = \varphi''(-\lambda/2) = -\frac{3}{4} \left(\frac{4}{5}\right)^{5/2} \frac{W}{\lambda^2} = -\frac{24}{25\sqrt{5}} \frac{W}{\lambda^2} \approx -0.42933 \frac{W}{\lambda^2}. \quad (26)$$

Now embed the contact in an environment of effective stiffness k_{env} acting on the depth coordinate, with the far field held at $\bar{h}$:

$$U(h) = \varphi(h) + \frac{1}{2}k_{\text{env}}(h - \bar{h})^2. \quad (27)$$

Stationary points satisfy $\varphi'(h) + k_{\text{env}}(h - \bar{h}) = 0$, and this equation has multiple solutions for some $\bar{h}$ — equivalently, the branch $h(\bar{h})$ has folds — if and only if $U'' = \varphi'' + k_{\text{env}}$ changes sign somewhere. By (26),

$$\text{the contact is a two-state hysteron} \iff k_{\text{env}} < \kappa_{\text{crit}} = \frac{24}{25\sqrt{5}} \frac{W}{\lambda^2} \quad (28)$$

with the two fold points located at the roots of $\varphi''(h) = -k_{\text{env}}$, which for k_{env} slightly below threshold straddle $h = -\lambda/2$ and separate as k_{env} decreases. Below threshold the bond snaps in at the outer fold on approach and snaps out at the inner one on retraction, dissipating the enclosed area; above threshold the surroundings hold the unstable branch and the bond forms and breaks reversibly.

Condition (28) is the precise sense in which a bond is a hysteron in this model, and it is checkable per contact from the assembled stiffness matrix. Two consequences are worth recording as predictions. Hysteron density should increase as the network softens, so approaching unjamming from above should convert reversible bonds into snapping ones and shift response toward avalanche-dominated, brittle behaviour. And the threshold scales as W/λ^2 rather than as W or F_{pull} , so short-ranged strong adhesion is hysteretic where long-ranged adhesion of equal work is not. Both are consistent with the intended role of adhesion as the brittle–ductile axis, and both are falsifiable within the model.

12 Hessian structure of the combined law

Differentiating $E = \varphi(h_\varepsilon)$ twice and inserting (8) and (10),

$$\nabla E = \varphi'(h_\varepsilon) \nabla h_\varepsilon = -\varphi' \bar{n}, \quad \nabla^2 E = \varphi'' \bar{n} \bar{n}^\top - \frac{\varphi'}{\varepsilon} \text{Cov}_w(n). \quad (29)$$

The geometric term now carries the sign of $-\varphi'$ rather than being unconditionally softening. Three regimes follow, and because $\text{Cov}_w(n)$ is positive semidefinite with $\bar{n}$ in its kernel complement, they can be separated cleanly into the direction along $\bar{n}$ and the direction tangent to the level set.

For $h > h^*$ the contact is net repulsive, $\varphi' > 0$, and the transverse eigenvalue is negative: this is the geometric softening of Section 8, with magnitude φ'/ε times the normal spread. For

$h < 0$ the contact is net attractive with $\varphi' < 0$, so the transverse term is stabilizing, but $\varphi'' < 0$ and the longitudinal direction is unstable — the snap-in instability of Section 11. In between,

$$\nabla^2 E \succeq 0 \quad \Longleftrightarrow \quad 0 \leq h_\varepsilon \leq h^*, \quad (30)$$

verified numerically to be exactly this interval. At $h = h^*$ the geometric term vanishes identically and the block degenerates to the rank-one positive form $\varphi''(h^*) \bar{n} \bar{n}^\top$.

This is the practical payoff of the adhesive law beyond the physics. In the purely repulsive model the transverse softening scales as $kh^{3/2}/\varepsilon$ and is unbounded as $\varepsilon \rightarrow 0$, so shape fidelity and conditioning are in direct conflict. With adhesion the softening scales as $|\varphi'(h)|/\varepsilon$, and φ' vanishes at the depth where contacts actually sit in a relaxed packing. The conditioning penalty is therefore controlled by the *excess* depth $h - h^*$ under load rather than by h itself, and ε can be made as small as the shape demands. The remaining requirement is only $\varepsilon \gtrsim (h_{\max} - h^*)$ for the most compressed contacts, which at moderate confining stress is a far weaker condition than $\varepsilon \gtrsim h_{\max}$.

13 The interfacial generalization: adhesion as a mollified chord

For a point in a loop, W in (19) is legitimately a lumped per-bond parameter: a point has no interface, so there is nothing for an interfacial energy density to multiply, and the objection raised in Section 9 does not bite. For loop–loop contact it does bite, and the correct object is a surface energy. If γ is the adhesion energy per unit length of shared interface, the exact energy for bodies A and B is $-\gamma \mathcal{H}^1(\partial A \cap B)$, and the coarea formula writes this as a level-set integral,

$$\mathcal{H}^1(\partial A \cap B) = \int_{\mathbb{R}^2} \delta(h^A) \Theta(h^B) |\nabla h^A| dx, \quad (31)$$

whose mollification is obtained by replacing $\Theta \rightarrow \chi_\lambda$ and $\delta \rightarrow \chi'_\lambda$ — the same pair of Plummer kernels, at no new analytic cost. Collapsing the δ onto the boundary gives the form to implement,

$$E_{\text{adh}} = -\gamma \int_{\partial A} \chi_\lambda(h_\varepsilon^B(x)) d\ell(x) \quad (32)$$

the arclength of A 's boundary weighted by its smooth membership in B : a mollified contact chord. It is a one-dimensional integral, it is as smooth in configuration as h_ε^B , and it recovers the sharp chord as $\lambda, \varepsilon \rightarrow 0$. Crucially it scales correctly with contact type, giving $O(|e|)$ for face-on-face contact nearly independent of indentation and $O(\sqrt{\lambda|e|})$ for vertex-on-face, so the two contact types have genuinely different adhesive energies and the switch between them is the source of the debond/rebond bistability. Symmetrizing as $\frac{1}{2}(E_{AB} + E_{BA})$ removes the arbitrary choice of which body's boundary to integrate.

Equation (32) evaluates in closed form in the regime the parameter hierarchy already requires. When $\varepsilon \ll \lambda$ the softmin defining h_ε^B is dominated by a single face of B over all but $O(\varepsilon)$ of each stretch, so parametrize edge i of A as $x(u) = v_i + u\hat{t}_i$ and write $h_\varepsilon^B \simeq \ell_j^B(x(u)) = \alpha - \beta u$ with $\alpha = c_j^B - n_j^B \cdot v_i$ and $\beta = n_j^B \cdot \hat{t}_i$. The primitive of the Plummer sigmoid is elementary,

$$\Psi_\lambda(z) \equiv \int \chi_\lambda(z) dz = \frac{1}{2} \left(z + \sqrt{z^2 + \lambda^2} \right), \quad \Psi'_\lambda(z) = \chi_\lambda(z), \quad (33)$$

a softplus-like function with $\Psi_\lambda(z) \rightarrow z$ for $z \gg \lambda$ and $\Psi_\lambda(z) \rightarrow \lambda^2/4|z|$ for $z \ll -\lambda$, so each stretch contributes

$$\int_{u_0}^{u_1} \chi_\lambda(\alpha - \beta u) du = \frac{\Psi_\lambda(\alpha - \beta u_0) - \Psi_\lambda(\alpha - \beta u_1)}{\beta}. \quad (34)$$

No new master integrals appear, and in particular nothing of the $J_{\log}$, J_{atan} family is needed. Two implementation points attach to (34). The limit $\beta \rightarrow 0$ is not a corner case to be tolerated but the physically dominant configuration — edge of A parallel to a face of B is precisely face-on-face contact — and (34) is catastrophically cancelling there, so switch to the series $\chi_\lambda(\alpha) \Delta u - \frac{1}{2}\beta \chi'_\lambda(\alpha) \Delta u^2 + O(\beta^2)$ once $|\beta| \Delta u \lesssim \lambda$. And the breakpoints where the dominant face of B switches along an edge of A make the piecewise-affine approximation C^1 but not C^2 in configuration, with Hessian jumps of relative size $O(\varepsilon/\lambda)$ at events where a breakpoint crosses a vertex of A . Since $\varepsilon \ll \lambda$ is independently required, these are small and the closed form is the recommended path; the alternative of fixed-order Gauss–Legendre quadrature with the full softmin at the nodes is exactly smooth, because fixed nodes on a smoothly moving interval move smoothly, but it costs several evaluations of h_ε^B per edge pair and buys back only $O(\varepsilon/\lambda)$.

The practical instruction for the point–loop code is therefore to carry W as a per-contact scalar returned by a function that presently returns a constant and will later return γ times the chord of (32). Nothing in (29), in the vertex derivatives of Section 14, or in the hysteron analysis of Section 11 changes under that substitution, except that W acquires its own configuration dependence and contributes additional terms to the gradient and Hessian through the chain rule.

14 Vertex derivatives and exact conservation

The loop’s degrees of freedom are its vertices, and the energy must be differentiated with respect to them, including through the normals. Let δ denote variation. From $\hat{t}_i = e_i/|e_i|$, in two dimensions $(I - \hat{t}_i \hat{t}_i^\top) \delta e_i = (n_i \cdot \delta e_i) n_i$, so

$$\delta \hat{t}_i = \frac{n_i \cdot \delta e_i}{|e_i|} n_i, \quad \delta n_i = J \delta \hat{t}_i = \frac{n_i \cdot \delta e_i}{|e_i|} J n_i = -\frac{n_i \cdot \delta e_i}{|e_i|} \hat{t}_i. \quad (35)$$

Define the normalized foot-of-perpendicular parameter

$$s_i(x) = \frac{\hat{t}_i \cdot (x - v_i)}{|e_i|}, \quad \text{so} \quad \hat{t}_i \cdot (v_i - x) = -s_i |e_i|. \quad (36)$$

Then from $\ell_i = n_i \cdot (v_i - x)$ and $\delta e_i = \delta v_{i+1} - \delta v_i$,

$$\delta \ell_i = \delta n_i \cdot (v_i - x) + n_i \cdot \delta v_i = s_i n_i \cdot (\delta v_{i+1} - \delta v_i) + n_i \cdot \delta v_i, \quad (37)$$

that is

$$\frac{\partial \ell_i}{\partial v_i} = (1 - s_i) n_i, \quad \frac{\partial \ell_i}{\partial v_{i+1}} = s_i n_i, \quad \frac{\partial \ell_i}{\partial x} = -n_i. \quad (38)$$

The normal load is distributed to the edge’s two endpoints with barycentric weights determined by where the perpendicular from x meets the edge line — the natural and correct lever rule. Note that $s_i \notin [0, 1]$ is possible, when the foot lies beyond an endpoint; one weight is then negative

and the distribution is an extrapolation, still exactly conservative, and in practice suppressed by a small w_i .

Summing (38) over all arguments gives $(1 - s_i)n_i + s_i n_i - n_i = 0$, so translating the point and the whole loop together leaves every ℓ_i and hence h_ε invariant: forces sum to zero identically, in floating point as well as in exact arithmetic. For rotations, the total torque contribution of edge i is

$$[(1 - s_i)v_i + s_i v_{i+1} - x] \times n_i = [v_i + s_i e_i - x] \times n_i = (q_i - x) \times n_i = \ell_i n_i \times n_i = 0, \quad (39)$$

using $v_i + s_i e_i = q_i$, the foot of the perpendicular, and $q_i - x = \ell_i n_i$ from the proof of Lemma 1. Torque balance is exact, edge by edge, with no cancellation between terms. Requirement (v) holds.

Assembling, the gradient with respect to vertex j picks up the two incident edges,

$$\frac{\partial h_\varepsilon}{\partial v_j} = w_{j-1} s_{j-1} n_{j-1} + w_j (1 - s_j) n_j, \quad \frac{\partial E}{\partial v_j} = \varphi'(h_\varepsilon) \frac{\partial h_\varepsilon}{\partial v_j}, \quad (40)$$

and second derivatives follow from (9) together with the second variation of n_i , which involves only $\hat{t}_i$, n_i , and $|e_i|^{-1}$; every entry remains algebraic.

15 Nonconvex loops

Lemma 1 fails for nonconvex Ω : the line supporting an edge adjacent to a reflex vertex cuts through the interior, so $\min_i \ell_i$ becomes negative at interior points and $\Omega \neq \bigcap_i \{\ell_i > 0\}$. The remedy is composition. Decompose $\Omega = \bigcup_k \Omega^{(k)}$ into convex pieces, build $h_\varepsilon^{(k)}$ for each by (5), and combine with the softmax

$$h_\varepsilon = +\varepsilon \log \sum_k e^{+h_\varepsilon^{(k)}/\varepsilon}, \quad (41)$$

which is again real-analytic, with gradient and Hessian following the same weighted-mean and weighted-covariance structure as (8)–(10) but with $+1/\varepsilon$ in the Hessian, so the combined operator is no longer sign-definite. This is the Rvachev **R-function** construction; it remains analytic under arbitrary nesting of unions, intersections and complements, at the cost of an $O(\varepsilon)$ error near every seam of the decomposition, including artificial seams internal to Ω . Reflex corners are rounded *outward* by $O(\varepsilon)$, the mirror of (15), so ε must be small against the smallest feature adjoining any reflex vertex. Identity (12) no longer holds exactly for the composed function, which removes the cheapest verification test; substitute finite-difference checks there.

16 Complementarity of the two mollifiers

The construction uses two distinct regularizations and it is worth stating plainly why neither can do the other's job, since both are analytic and both have closed forms of comparable elegance.

The mollified-indicator route produces a smooth membership function. A regularized winding number assigns to each edge a contribution

$$\frac{h}{2\pi\sqrt{h^2 + \lambda^2}} \left[\arctan \frac{(1-s)|e|}{\sqrt{h^2 + \lambda^2}} + \arctan \frac{s|e|}{\sqrt{h^2 + \lambda^2}} \right], \quad (42)$$

summing to a function that tends to 1 inside and 0 outside; against a straight edge at signed offset δ the profile collapses to $\delta/\sqrt{\delta^2 + \lambda^2}$, which is $2\chi_\lambda(\delta) - 1$. This *saturates* for $|\delta| \gg \lambda$. As a depth it is therefore useless: it resolves the boundary at scale λ and carries no information beyond it, so a force law built on it has a range set by the regularizer rather than by the particle geometry, and its gradient is $O(1/\lambda)$ at contact rather than $O(1)$.

The log-sum-exp route produces a metric. It does not saturate; by (11) its gradient has unit magnitude to exponential accuracy wherever a single face dominates, which is exactly the property a depth must have and the one a membership function must not. As an adhesion it would be equally useless, since an interaction whose strength grew linearly and without bound with engagement would have infinite work of separation.

The two failures are mirror images, and the resolution is to use each where its defect is a virtue: log-sum-exp for h_ε , whose job is to measure, and Plummer for χ_λ , whose job is to switch. They share the algebraic primitives ℓ_i , n_i and s_i , so the fused kernel evaluates both from one pass over the edge list, and the identity $\Psi'_\lambda = \chi_\lambda$ of (33) extends the same family to the interfacial energy of Section 13 without introducing a third regularization or a new special function.

17 Numerics, parameter selection, validation

Evaluate (5) in shifted form. With $m = \min_i \ell_i$,

$$h_\varepsilon = m - \varepsilon \log \sum_i e^{-(\ell_i - m)/\varepsilon}, \quad (43)$$

in which every exponent is ≤ 0 and the sum lies in $[1, N]$; the unshifted form overflows as soon as ℓ_i/ε exceeds a few hundred, which for $\varepsilon \sim 10^{-3}$ in units of the particle size means immediately. The weights w_i are computed from the same shifted exponentials.

Cost is $O(N)$ per point-loop pair. On the GPU it is tempting to truncate to the edges with $\ell_i - m < \varepsilon\Lambda$, which costs a relative error $e^{-\Lambda}$ in the sum and $\varepsilon e^{-\Lambda}$ in h_ε ; with $\Lambda \approx 30$ this is below double-precision resolution of anything else in the calculation. Strictly, truncation reintroduces a discontinuity, so the active set should be frozen for the duration of a Newton solve and rebuilt between steps, or the cutoff replaced by a smooth window; in practice the jump is far below solver tolerance and freezing suffices.

The adhesive term needs no shifted evaluation, being a bounded rational-algebraic expression, but it does have an algebraic tail: $1 - \chi_\lambda(h) \sim \lambda^2/4h^2$ for $h \ll -\lambda$, so the interaction is formally infinite-ranged. Truncating the pair list at $h_c = -n\lambda$ omits an energy $O(W/4n^2)$ and leaves a force discontinuity $O(W/2n^3\lambda)$; at $n = 10$ these are $2.5 \times 10^{-3}W$ and $5 \times 10^{-4}W/\lambda$ respectively, below any sensible solver tolerance but nonzero, and the truncation should be recorded as a stated approximation rather than absorbed silently into the neighbour-list radius.

Parameter hierarchy. The model carries four parameters, k , ε , W , λ , of which k sets the stress unit and μ of (24) is the dimensionless combination to sweep. The constraints on the two lengths are as follows. The rounding must be finer than the adhesive range, $\varepsilon \ll \lambda$, or adhesion samples the corner rounding and the bond strength at a corner becomes an artifact of the regularizer rather than a property of the shape; this is also what licenses the closed form (34). The adhesive range must be short against the shortest edge, $\lambda \ll \min_i |e_i|$, or the chord in (32) is smeared across several faces and the face/vertex contact distinction is lost. The equilibrium

indentation must be small against the inradius, $h^* = \lambda \eta(\mu) \ll r_{\text{in}}$, or the point equilibrates near the medial axis where (11) has degraded; by the large- μ asymptote this bounds $\mu \ll (r_{\text{in}}/\lambda)^3$, which for $\lambda/r_{\text{in}} \sim 5 \times 10^{-2}$ is $\mu \ll 10^4$ and is not a practical restriction. Finally $\varepsilon \gtrsim h_{\text{max}} - h^*$ for conditioning, per Section 12. Note that these do *not* force $h^* \lesssim \varepsilon$: an earlier reading of the Section 8 crossover suggested that they did, which would have confined the model to $\mu \ll 1$ and made the JKR end of the sweep inaccessible. The correct constraint involves the excess depth beyond equilibrium, and the full range of μ is available. With $\varepsilon \sim 10^{-3}$ and $\lambda \sim 10^{-2}$ in units of the particle diameter there is roughly a decade of headroom in each inequality.

Validation. Seven independent checks, in order of increasing cost. First, (12): evaluate $|\nabla h_\varepsilon|^2 - \varepsilon \Delta h_\varepsilon - 1$ at random points and confirm it is at machine precision. This tests the gradient and the Hessian trace simultaneously against an exact identity and needs no reference solution. Second, invariance: confirm $\sum_j \partial_{v_j} h_\varepsilon + \partial_x h_\varepsilon = 0$ and the torque sum of Section 14, again to machine precision. Third, Proposition 1: confirm $0 \leq \min_i \ell_i - h_\varepsilon \leq \varepsilon \log N$ everywhere and that the deficit decays as $\varepsilon(N-1)e^{-\Delta/\varepsilon}$ along a path leaving the medial axis. Fourth, the equilibrium cubic: solve $\varphi'(h^*) = 0$ numerically and confirm $\eta^3 + \eta = \mu$ and the Cardano expression agree to solver precision over several decades in μ . Fifth, the work of adhesion: quadrature of $W\chi'_\lambda$ over the line must return W , and $\min_h \varphi''$ must occur at $h = -\lambda/2$ with the value (26). Sixth, the semidefiniteness window (30): scan h and confirm the smallest eigenvalue of (29) is nonnegative on exactly $[0, h^*]$, with the geometric term vanishing at h^* . Seventh, complex-step or high-order finite differences on $\partial E/\partial x$ and $\partial E/\partial v_j$ against (29) and Section 14, the only check that exercises the vertex second derivatives and the only one needing care with the $h_+^{1/2}$ factor near contact onset. For the loop-loop extension, add an eighth: (32) divided by $-\gamma$ must converge to the analytically known chord length as $\lambda, \varepsilon \rightarrow 0$ in a fixed face-on-face configuration, and (34) must agree with its small- β series across the switchover.